NUCLEAR POWER REACTORS

Molten Salt Reactors

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- Molten salt reactors operated in the 1960s.
- They are seen as a promising technology today principally as a thorium fuel cycle prospect or for using spent LWR fuel.
- A variety of designs is being developed, some as fast neutron types.
- Global research is currently led by China.
- Some have solid fuel similar to HTR fuel, others have fuel dissolved in the molten salt coolant.

Molten salt reactors (MSRs) use molten fluoride salts as primary coolant, at low pressure. This itself is not a radical departure when the fuel is solid and fixed. But extending the concept to dissolving the fissile and fertile fuel in the salt certainly represents a leap in lateral thinking relative to nearly every reactor operated so far. However, the concept is not new, as outlined below.

MSRs may operate with epithermal or fast neutron spectrums, and with a variety of fuels. Much of the interest today in reviving the MSR concept relates to using thorium (to breed fissile uranium-233), where an initial source of fissile material such as plutonium-239 needs to be provided. There are a number of different MSR design concepts, and a number of interesting challenges in the commercialisation of many, especially with thorium.

Privacy - Terms

The salts concerned as primary coolant, mostly lithium-beryllium fluoride and lithium fluoride, remain liquid without pressurization from about 500°C up to about 1400°C, in marked contrast to a PWR which operates at about 315°C under 150 atmospheres pressure.

The main MSR concept is to have the fuel dissolved in the coolant as fuel salt, and ultimately to reprocess that online. Thorium, uranium, and plutonium all form suitable fluoride salts that readily dissolve in the LiF-BeF_2 (FLiBe) mixture, and thorium and uranium can be easily separated from one another in fluoride form. Batch reprocessing is likely in the short term, and fuel life is quoted at 4-7 years, with high burn-up. Intermediate designs and the AHTR have fuel particles in solid graphite and have less potential for thorium use.

Graphite as moderator is chemically compatible with the fluoride salts.

Background

The US MSR programme originated in the Aircraft Reactor Experiment* at the Oak Ridge National Laboratory (ORNL), Tennessee (built as part of the wartime Manhattan Project).

* It had primary coolant of NaF-ZrF_4 with UF_4 and secondary coolant of NaK at 880°C, to produce 60 MWt.

During the 1960s, the USA developed the molten salt breeder reactor concept at the ORNL. It was the primary back-up option for the fast breeder reactor (cooled by liquid metal) and the small prototype 8 MWt Molten Salt Reactor Experiment (MSRE) operated at Oak Ridge over four years to 1969 (the MSR programme ran 1957-1976). In the first campaign (1965-68), uranium-235 tetrafluoride (UF_4) enriched to 33% was dissolved in molten lithium, beryllium and zirconium fluorides at 600-650°C which flowed through a graphite moderator at ambient pressure. The fuel comprised about one percent of the fluid.*

* LiF-BeF_2-ZrF_4-UF_4 (65:29.1:5:0.9 mole %)

The coolant salt in a secondary circuit was eutectic lithium + beryllium fluoride (FLiBe).* There was no breeding blanket, this being omitted for simplicity in favour of neutron measurements.

* Fuel salt melting point 434°C, coolant salt melting point 455°C. See Wong & Merrill 2004 reference.

The original objectives of the MSRE to demonstrate the concept as a simple reliable reactor were achieved by March 1965, and the U-235 campaign concluded. A second campaign (1968-69) used U-233 fuel which was then available, making MSRE the first reactor to use U-233, though it was imported and not bred in the reactor. This programme prepared the way for building a MSR breeder utilizing thorium, which would operate in the thermal (slow) neutron spectrum. A total of 10,000 operating hours was logged.

The culmination of the Oak Ridge research over 1970-76 resulted in an MSR design that would use $LiF-BeF_2-ThF_4-UF_4$ (72:16:12:0.4) as primary coolant with fuel. It would be moderated by graphite with a four-year replacement schedule, use $NaF-NaBF_4$ as the secondary coolant, and have a peak operating temperature of 705°C.[1]

The R&D programme demonstrated the feasibility of this system, albeit excluding online reprocessing, and highlighted some unique corrosion and safety issues that would need to be addressed if constructing a larger pilot MSR with fuel salt. With a radioactive primary coolant loop, challenges would include processing facilities to remove the main fission products, though gaseous fission products come off readily in the gas purge system. It also showed that breeding required a different design, with a larger blanket loop and two fluids (heterogeneous). Tritium production was a problem (see below re lithium enrichment). The breeder project to produce U-233 from thorium was discontinued.

From 1975 to 2010 little happened in the USA with MSRs, though in 1980 ONRL published a study to "examine the conceptual feasibility" of a denatured MSR (DMSR) fuelled with low-enriched uranium-235 "and operated with a minimum of chemical processing," solely as a burner reactor. The main priority was proliferation resistance, avoiding use of HEU.[2]

In the UK a large (2.5 GWe) lead-cooled fast spectrum MSR (MSFR) with the plutonium fuel dissolved in a molten chloride salt was designed, with experimental work undertaken over 1968-73. Funding ceased in 1974.

There is now renewed interest in the MSR concept in Japan, Russia, China, France and the USA, and one of the six Generation IV designs selected for further development is the MSR in two distinct variants, the molten salt fast reactor (MSFR) and the advanced high temperature reactor (AHTR) – also known as the fluoride salt-cooled high-temperature reactor (FHR) with solid fuel, or PB-FHR specifically with pebble fuel. The Generation IV international Forum

(GIF) mentions 'salt processing' as a technology gap for MSRs, putting the initial focus clearly on burners rather than breeders.

Since the 2002 Generation IV selection process, significant changes in design philosophy have taken place, according to a 2015 report by Energy Process Developments Ltd (EPD). The first is to design simpler, less ambitious, MSRs that do not breed new fuel, do not require online fuel reprocessing and which use the well-established enriched uranium fuel cycle. In this regard, both American researchers and the China Academy of Sciences/SINAP are working on solid fuel, salt-cooled reactor technology as a realistic first step into MSRs. In 2014, as part of an assessment of MSR activity internationally, proposals were made for pilot-scale implementation, where technical readiness was claimed. Six such specific proposals* were assessed over 12 months with commissioned expertise from established UK nuclear engineering firms. These proposals were all seen as credible for building a prototype, with one emerging in the EPD report as currently most suitable as a basis for UK MSR development, the Moltex SSR.

* from Flibe Energy, ThorCon, Moltex, Seaborg Technologies, Terrestrial Energy and Transatomic Power.

Function

In the normal or basic MSR concept, the fuel is a molten mixture of lithium and beryllium fluoride (FLiBe) salts with dissolved low-enriched uranium (U-235 or U-233) fluorides (UF_4). The core consists of unclad graphite moderator arranged to allow the flow of salt at about 700°C and at low pressure. Much higher temperatures are possible but not yet tested. Heat is transferred to a secondary salt circuit and thence to steam or process heat. The basic design is not a fast neutron reactor, but with some moderation by the graphite is epithermal (intermediate neutron speed) and breeding ratio is less than 1.

However, this concept, with fuel dissolved in the salt, is further from commercialisation than solid fuel designs, where the ceramic fuel may be set in prisms, plates, or pebbles, or one design with liquid fuel in static tubes. Reprocessing that fuel salt online is even further from commercialization.

Considering liquid-fuel MSR designs, thorium can be dissolved with the uranium in a single fluid MSR, known as a homogeneous design. Two-fluid, or heterogeneous MSRs, would have fertile salt containing thorium in a second loop separate from the fuel salt

containing fissile uranium or plutonium and could operate as a breeder reactor (MSBR). Here, the U-233 is progressively removed* and transferred to the primary circuit. However, graphite degradation from neutron flux limits the useful life of the reactor core with the fuel and breeding fluids in close juxtaposition, and in the 1960s it was assumed that the entire reactor vessel in the two-fluid design would be replaced after about eight years.**

* e.g. by bubbling fluorine through the salt so that UF_6 is formed and removed as a gas. The UF_6 is reduced and added to the fuel stream.

** Graphite is used to slow neutrons in epithermal designs, and deteriorates in a high neutron flux environment. The rate of damage increases with temperature, which is a particular problem with MSRs at 700°C.

In liquid-fuel MSR designs the fission products dissolve in the fuel salt and are ideally removed continuously in an adjacent online reprocessing loop and replaced with fissile uranium, plutonium and other actinides or, potentially, fertile Th-232 or U-238. Meanwhile caesium and iodine in particular remain secure in the molten salt. Xenon is removed rapidly by outgassing, but protactinium-233 is a problem with thorium as a fuel source. (It is an intermediate product in producing U-233 and is a major neutron absorber.) Constant removal of fission products means that a much higher fuel burn-up could be achieved (> 50%) and the removal of fission products means less decay heat to contend with after reactor shutdown. Actinides are fully recycled and remain in the reactor until they fission or are converted to higher actinides which do so. Hence fissile plutonium is largely consumed, and contributes significant energy. The high-level waste would comprise fission products only, hence with shorter-lived radioactivity.

Compared with solid-fuelled reactors, MSR systems with circulating fuel salt are claimed to have lower fissile inventories*, no radiation damage constraint on fuel burn-up, no requirement to fabricate and handle solid fuel or solid used fuel, and a homogeneous isotopic composition of fuel in the reactor. Actinides are less-readily formed from U-233 than in fuel with atomic mass greater than 235. These and other characteristics may enable MSRs to have unique capabilities and competitive economics for actinide burning and extending fuel resources. Safety is high due to passive cooling up to any size. Also, several designs have freeze plugs so that if excessive temperatures are reached, the primary salt will be drained by gravity away from the moderator into dump tanks configured to prevent criticality.

* In particular, a small inventory of weapons-fissile material (Pu-242 being the dominant Pu isotope remaining), and low fuel use (the French self-breeding variant claims 50kg of thorium and 50kg U-238 per billion kWh).

MSRs have large negative temperature and void coefficients of reactivity, and are designed to shut down due to expansion of the fuel salt as temperature increases beyond design limits. The negative temperature and void reactivity coefficients passively reduce the rate of power increase in the case of an inadvertent control rod withdrawal (technically known as a 'reactivity insertion'). When tests were made on the MSRE, a control rod was intentionally withdrawn during normal reactor operations at full power (8 MWt) to observe the dynamic response of core power. Such was the rate of fuel salt thermal expansion that reactor power levelled off at 9 MWt without any operator intervention.

The MSR thus has a significant load-following capability where reduced heat abstraction through the boiler tubes leads to increased coolant temperature, or greater heat removal reduces coolant temperature and increases reactivity. Primary reactivity control is using the secondary coolant salt pump or circulation which changes the temperature of the fuel salt in the core, thus altering reactivity due to its strong negative reactivity coefficient. The MSR works at near atmospheric pressure, eliminating the risk of explosive release of volatile radioactive materials.

One MSR developer, Moltex, has put forward a molten salt heat storage concept (GridReserve) to enable the reactor to supplement intermittent renewables. Hot nitrate salt at about 600°C is transferred to storage tanks which are able to hold eight hours of reactor output at 2.5 GW thermal (as used in solar CSP plants). The heat store is said to add only £3/MWh to the levelised cost of electricity.

In the MSBR, the reactor-grade U-233 bred in the secondary circuit needs to be removed, or it will fission there and contaminate that circuit with 'hot' fission products. Therefore in practice the protactinium (Pa-233) formed from the thorium needs to be removed before it decays to U-233*, but this process is unproven at any scale. It is relatively easy to remove the U-233 from the Pa-233 by fluorination to UF_6 before reducing it to UF_4 for adding to the primary fuel salt circuit. However, the U-233 is contaminated with up to 400 ppm U-232 which complicates processing, due to its highly gamma-active decay progeny.

* Th-232 gains a neutron to form Th-233, which soon beta decays (half-life 22 minutes) to protactinium-233. The Pa-233 (half-life of 27 days) decays

into U-233. Some U-232 is also formed via Pa-232 along with Th-233, and a decay product of this is very gamma active.

MSRs would normally operate at much higher temperatures than LWRs – up to at least 700°C, and hence have potential for process heat. Up to this temperature, satisfactory structural materials are available. 'Alloy N' is a nickel-based alloy (Ni-Cr-Mo-Si) developed at ORNL specifically for MSRs with fluoride salts.

Primary and secondary cooling, the fluoride salts

Fluoride salts have very low vapour pressure even at red heat, carry more heat than the same volume of water, have reasonably good heat transfer properties, are not damaged by radiation, do not react violently with air or water, and are inert to some common structural metals. However having the fuel in solution also means that the primary coolant salt becomes radioactive, complicating maintenance procedures, and the chemistry of the salt must be monitored closely to maintain a chemically reduced state to minimise corrosion. Also the beryllium in the salt is toxic, which leads to at least one design avoiding it, though this requires higher temperatures to keep LiF liquid. LiF however can carry a higher concentration of uranium than FLiBe, allowing less enrichment. There are difficulties with plutonium and other TRU fluorides in fluoride solvents.

Lithium used in the salt must be fairly pure Li-7, since Li-6 produces tritium when (readily) fissioned by neutrons. Li-7 has a very small neutron cross-section (0.045 barns). This means that lithium must be enriched beyond its natural 92.5% Li-7 level to minimise tritium production. Lithium-7 is being produced at least in Russia and possibly China today as a by-product of enriching lithium-6 to produce tritium for thermonuclear weapons. See also Lithium paper.

LiF is exceptionally stable chemically, and the LiF-BeF_2 mix ('FLiBe')* is eutectic (at 459°C it has a lower melting point than either ingredient – LiF is about 500°C). It boils at 1430°C. It is favoured in MSR and AHTR primary cooling and when uncontaminated has a low corrosion effect. The three nuclides (Li-7, Be, F) are among the few to have low enough thermal neutron capture cross-sections not to interfere with fission reactions.

* Approx. 2:1 molar, hence sometimes represented as Li_2BeF_4.

LiF without the toxic beryllium solidifies at about 500°C and boils at about 1200°C. FLiNaK (LiF-NaF-KF) is also eutectic and solidifies at 454°C and boils at 1570°C. It has a higher neutron cross-section than FLiBe or LiF but can be used in intermediate cooling loops. Sodium-beryllium fluoride (BeF_2-NaF) solidifying at 385°C is used as fuel salt in one design for cost reasons.

The hot molten salt in the primary circuit can be used with secondary salt circuit or secondary helium coolant generating power via the Brayton cycle as with HTR designs, with potential thermal efficiencies of 48% at 750°C to 59% at 1000°C, or simply with steam generators. In industrial applications molten fluoride salts (possibly simply cryolite – Na-Al fluoride) are a preferred interface fluid in a secondary circuit between the nuclear heat source and any chemical plant. The aluminium smelting industry provides substantial experience in managing them safely.

Most secondary coolant salts do not use lithium, for cost reasons. ZrF_4-NaF-KF, ZrF_4-KF, NaF-BeF_2 eutectic mixes are usual, as well as LiF-NaF-KF (FLiNaK).

In the secondary cooling circuit of the AHTR concept, air is compressed, heated, flows through gas turbines producing electricity, enters a steam recovery boiler producing steam that produces additional electricity, and exits to the atmosphere. Added peak power can be produced by injecting natural gas (or hydrogen in the future) after nuclear heating of the compressed air to raise gas temperatures and plant output, giving it rapidly variable output (of great value in grid stability and for peak load demand where renewables have significant input). This is described as an air Brayton combined-cycle (ABCC) system in secondary circuit.

In the 1960s MSRE, an alternative secondary coolant salt considered was 8% NaF + 92% NaF-BeF_2 with melting point 385°C, though this would be more corrosive.

Chloride salts, fast spectrum reactors

Chloride salts have some attractive features compared with fluorides, in particular the actinide trichlorides form lower melting point solutions and have higher solubility for actinides so can contain significant amounts of transuranic elements. $PuCl_3$ in NaCl has been well researched. While NaCl has good nuclear, chemical and physical properties, its high melting point means it needs to be blended with $MgCl_2$ or $CaCl_2$, the former being preferred in eutectic, and allowing the addition of actinide trichlorides. The major isotope

of chlorine, Cl-35 gives rise to Cl-36 as an activation product – a long-lived energetic beta source, so Cl-37 is much preferable in a reactor.

A British design contains the chloride fuel salt in vertical tubes and relies on convection to circulate the secondary salt coolant, which is a fluoride mix.

Fast spectrum MSRs (MSFRs) can have conversion ratios ranging from burner to converter to breeder. This may be within a single unit as the ratio of U-238 to transuranics (TRU) is varied – less U-238 giving more fission. They can be optimised for burning minor actinides, for breeding plutonium from U-238, or they may be open-cycle power plants without heavy metal separation from fission products. The fast neutron spectrum allows the possibility of not having onsite processing to remove TRUs. While fission products have relatively large neutron capture cross sections in the thermal energy range, the capture cross sections at higher energies is much lower, allowing much greater fission product build-up in an MSFR than in a thermal-spectrum MSR (gaseous fission products separate out of the liquid fuel). Eventually the fuel salt heavily loaded with fission products can be sent occasionally for batch processing or allowed to solidify and be disposed of in a repository. For full breeder configuration the fissile material needs to be progressively removed.

MSFRs have a negative void coefficient in the salt and a negative thermal reactivity feedback, so can maintain a high power density with passive safety. Freeze plugs to drain the fuel salt are a further passive safety measure as in other MSRs.

MSR research emphasis

American researchers and the China Academy of Sciences/SINAP are working primarily on solid fuel MSR technology. The main reason is that this is a realistic first step. In China this is focused on thorium-fuelled versions (see TMSR in *China's dual programme* section below). The technical difficulty of using molten salts is significantly lower when they do not have the very high activity levels associated with them bearing the dissolved fuels and wastes. The experience gained with component design, operation, and maintenance with clean salts makes it much easier then to move on and consider the use of liquid fuels, while gaining several key advantages from the ability to operate reactors at low pressure and deliver higher temperatures.

In the Generation IV programme for the MSR, collaborative R&D is pursued by interested members under the auspices of a provisional steering committee. There will be a long lead time to prototypes, and the R&D orientation has changed since the project was set up, due to increased interest. It now has two baseline concepts:

- The Molten Salt Fast Neutron Reactor (MSFR), which will take in thorium fuel cycle, recycling of actinides, closed Th/U fuel cycle with no U enrichment, with enhanced safety and minimal wastes. it is a liquid-fuel design.

- The Advanced High-Temperature Reactor (AHTR) – also known as the fluoride salt-cooled high-temperature reactor (FHR) – with the same graphite and solid fuel core structures as the VHTR and molten salt as coolant instead of helium, enabling power densities 4 to 6 times greater than HTRs and power levels up to 4000 MWt with passive safety systems. A 5 MWt prototype is under construction at Shanghai Institute of Nuclear Applied Physics (SINAP, under the China Academy of Sciences).

The GIF 2014 Roadmap said that a lot of work needed to be done on salts before demonstration reactors were operational, and suggested 2025 as the end of the viability R&D phase.

Russia's Molten Salt Actinide Recycler and Transmuter (MOSART) is a fast reactor fuelled only by transuranic (TRU) fluorides from uranium and MOX LWR used fuel. It is part of the MARS project (minor actinide recycling in molten salt) involving RIAR, Kurchatov and other research organisations. The 2400 MWt design has a homogeneous core of Li-Na-Be or Li-Be fluorides without a graphite moderator and has reduced reprocessing compared with the original US design. Thorium may also be used, though it is described as a burner-converter rather than a breeder.

The SAMOFAR (Safety Assessment of the Molten Salt Fast Reactor) project, based in the Netherlands and funded by the European Commission, aims to prove the safety concepts of the MSFR in breeding mode from thorium. It plans advanced experimental and numerical techniques, to deliver a breakthrough in nuclear safety and optimal waste management, and to create a consortium of stakeholders. "The use of the Th-U fuel cycle is of particular interest to the MSR, because this reactor is the only one in which the Pa-233 can be stored in a hold-up tank to let it decay to U-233." The SAMOFAR consortium consists of 11 participants and is mainly undertaken by universities and research laboratories such as CNRS,

JRC, CIRTEN, TU Delft and PSI, thereby exploiting each other's expertise and infrastructure. It commenced in 2015.

China's dual programme

China plans for the TMSR-SF to be an energy solution for the northwest half of the country, with lower population density and little water. The application of water-free cooling in arid regions is envisaged from about 2025.

The China Academy of Sciences in January 2011 launched an R&D programme on LFTRs, known there as the thorium-breeding molten-salt reactor (Th-MSR or TMSR), and claimed to have the world's largest national effort on it, hoping to obtain full intellectual property rights on the technology. This is also known as the fluoride salt-cooled high-temperature reactor (FHR). The TMSR Centre at Shanghai Institute of Nuclear Applied Physics (SINAP, under the China Academy of Sciences) at Jiading is responsible. In the 1970s SINAP worked towards building a 25 MWe MSR, but this endeavour gave way to the Qinshan PWR project.

SINAP has two streams of TMSR development – solid fuel (TRISO in pebbles or prisms/blocks) with once-through fuel cycle, and liquid fuel (dissolved in fluoride coolant) with reprocessing and recycle. A third stream of fast reactors to consume actinides from LWRs is planned. The aim is to develop both the thorium fuel cycle and non-electrical applications in a 20-30 year timeframe.

- The TMSR-SF stream has only partial utilization of thorium, relying on some breeding as with U-238, and needing fissile uranium input as well. It is optimized for high-temperature based hybrid nuclear energy applications. SINAP aimed at a 2 MW pilot plant initially, though this has been superseded by a simulator (TMSR-SF0). A 100 MWt demonstration pebble bed plant (TMSR-SF2) with open fuel cycle is planned by about 2025. TRISO particles will be with both low-enriched uranium and thorium, separately.

- The TMSR-LF stream claims full closed Th-U fuel cycle with breeding of U-233 and much better sustainability with thorium but greater technical difficulty. It is optimized for utilization of thorium with electrometallurgical pyroprocessing. SINAP aims for a 2 MWt pilot plant (TMSR-LF1) initially, then a 10 MWt experimental reactor (TMSR-LF2) by 2025, and a 100 MWt demonstration plant (TMSR-LF3) with full electrometallurgical

reprocessing by about 2035, followed by 1 a GW demonstration plant. The TMSR-LF timeline is about ten years behind the SF one.

- A TMSFR-LF fast reactor optimized for burning minor actinides is to follow.

SINAP sees molten salt fuel being superior to the TRISO fuel in effectively unlimited burn-up, less waste, and lower fabricating cost, but achieving lower temperatures (600°C+) than the TRISO fuel reactors (1200°C+). Near-term goals include preparing nuclear-grade ThF_4 and ThO_2 and testing them in a MSR. It appears that the postponement of building the 2 MW test reactor may be due to inadequate supplies of pure lithium-7.

The TMSR-SF programme is proceeding with preliminary engineering design in cooperation with the Nuclear Power Institute of China (NPIC) and Shanghai Nuclear Engineering Research & Design Institute (SNERDI). Nickel-based alloys are being developed for structures, along with very fine-grained graphite.

Two methods of tritium stripping are being evaluated, and also tritium storage.

The TMSR-SF0 is one-third scale and has a 370 kW electric heat source with FLiNaK primary coolant at 650°C and FLiNaK secondary coolant.

The 10 MWt TMSR-SF1 has 17% enriched TRISO fuel in 60mm pebbles, similar to HTR-PM fuel, and coolant at 630°C and low pressure. Primary coolant is FLiBe (with 99.99% Li-7) and secondary coolant is FLiNaK. Core height is 3 m, diameter 2.85 m, in a 7.8 m high and 3 m diameter pressure vessel. Residual heat removal is passive, by cavity cooling. A 20-year operating life was envisaged but the project is discontinued.

The 2 MWt TMSR-LF1 is under construction at Wu Wei in Gansu in a $3.3 billion programme. It will use fuel enriched to under 20% U-235, have a thorium inventory of about 50 kg and conversion ratio of about 0.1. FLiBe with 99.95% Li-7 would be used, and fuel as UF_4. The project would start on a batch basis with some online refuelling and removal of gaseous fission products, but discharging all fuel salt after 5-8 years for reprocessing and separation of fission products and minor actinides for storage. It would proceed to a continuous process of recycling salt, uranium and thorium, with online separation of fission products and minor actinides. It would work up from about 20% thorium fission to about 80%.

Beyond these, a 373 MWt/168 MWe liquid-fuel MSR small modular reactor is planned, with supercritical CO_2 cycle in a tertiary loop at 23 MPa using Brayton cycle, after a radioactive isolation secondary loop. Various applications as well as electricity generation are envisaged. It would be loaded with 15.7 tonnes of thorium and 2.1 tonnes of uranium (19.75% enriched), with one kilogram of uranium added daily, and have 330 GWd/t burn-up with 30% of energy from thorium. Online refuelling would enable eight years of operation before shutdown, with the graphite moderator needing attention.

The US Department of Energy is collaborating with the China Academy of Sciences on the programme, which had a start-up budget of $350 million. TMSR commercial deployment is anticipated in the 2030s.

Other solid- or fixed-fuel types

AHTR / FHR

Research on molten salt coolant has been revived at Oak Ridge National Laboratory (ORNL) in the USA with the Advanced High Temperature Reactor (AHTR). This is a larger reactor using a coated-particle graphite-matrix TRISO fuel like that in the GT-MHR (see the information paper on Small Nuclear Power Reactors) and with molten fluoride (FLiBe) salt as primary coolant. It is also known as the Fluoride High Temperature Reactor (FHR). While similar to the gas-cooled HTR it operates at low pressure (less than 1 atmosphere) and higher temperature, and gives better heat transfer than helium. The FLiBe salt is used solely as coolant, and achieves temperatures of 750-1000°C or more while at low pressure. This could be used in thermochemical hydrogen manufacture.

A small version of the AHTR/FHR is the SmAHTR, with 125 MWt size-matched to early process heat markets, or producing 50+MWe. The operating temperature is 700°C with FLiBe primary coolant and three integral heat exchangers. It is truck transportable, being 9m long and 3.5m diameter. Fuel is 19.75% enriched uranium in TRISO particles in graphite blocks or fuel plates. The refuelling interval is 2.5 to 4 years depending on fuel configuration. Secondary coolant is FLiNaK to Brayton cycle, and for passive decay heat removal, separate auxiliary loops go to air-cooled radiators. Later versions are intended to reach 850° to 1000°C, using materials yet to be developed.

AHTR reactor sizes of 1500 MWe/3600 MWt are envisaged, with capital costs estimated at less than $1000/kW.

In the secondary cooling circuit, air is compressed, heated, flows through gas turbines producing electricity, enters a steam recovery boiler producing steam that produces additional electricity, and exits to the atmosphere. Added peak power can be produced by injecting natural gas (or hydrogen in the future) after nuclear heating of the compressed air to raise gas temperatures and plant output, giving it rapidly variable output (of great value in grid stability and for peak load demand where renewables have significant input). This is described as an air Brayton combined-cycle (ABCC) system in secondary circuit.

Kairos Power FHR and Hermes

Kairos Power in the USA has designed a 320 MWt/140 MWe fluoride (FLiBe) salt-cooled high temperature reactor (KP-FHR) which it plans to build at the East Tennessee Technology Park at Oak Ridge, Tennessee, in collaboration with Oak Ridge National Laboratory. The reactor uses 19.75% enriched TRISO fuel in pebble form with online refuelling and operates at up to 650°C. Secondary circuit salt is 'solar' nitrate, feeding a steam generator ('60/40 nitrate salt', *i.e.* 60:40 by weight sodium nitrate to potassium nitrate). It has passive shutdown and decay heat removal. The prototype is the Hermes reduced-scale test reactor (<50 MWt), selected by the DOE in December 2020 for a $629 million programme over seven years (DOE share $303 million from the Advanced Reactor Demonstration Program). In May 2021 the Tennessee Valley Authority (TVA) agreed to provide engineering, operations, and licensing support for the Hermes project. TVA holds an early site permit for the Clinch River site.

Moltex SSR

Moltex Energy's Stable Salt Reactor (SSR) is a conceptual UK reactor design that, like all conventional reactors in operation, relies on convection from static vertical fuel tubes in the core to convey heat to the reactor coolant. Because the nuclear material is contained in fuel assemblies, standard industrial pumps can be used for the low radioactivity coolant salt. Core temperature is 500-600°C, at atmospheric pressure. Decay heat is removed by natural air convection.

Fuel tubes of nickel-chromium alloy three-quarters filled with the molten fuel salt (60% NaCl, 40% Pu, U & lanthanide trichlorides) are grouped into fuel assemblies which are similar to those used in standard reactors and use similar structural materials. The individual fuel tubes are vented so that fission product gases escape into the coolant salt, which is a NaF-KF-ZrF_4 mix (Li-7 fluoride is avoided for cost reasons). The assemblies can be moved laterally without removing them. Refuelling is thus continuous online, and after five years depleted assemblies are stored at one side of the pool pending reprocessing.

SSR factory-produced modules are 150 MWe containing fuel, pumps, primary heat exchanger, control blades and instrumentation. Several, up to gigawatt-scale, can share a reactor tank, half-filled with the coolant salt which transfers heat away from the fuel assemblies to the peripheral steam generators, essentially by convection, at atmospheric pressure. There are three versions: the SSR-W (Stable Salt Reactor – Wasteburner); the SSR-U (uranium-fuelled); and the SSR-Th (thorium).

The SSR-W is simplest and cheapest, due to compact core and no moderator. The primary fissile fuel in this original fast reactor version was to be plutonium-239 chloride with minor actinides and lanthanides, recovered from LWR fuel or from an SSR-U reactor. In 2020 the SSR-W fuel was 25% reactor-grade $PuCl_3$ with 30% UCl_3 and 45% KCl. This is in 451 fuel assemblies in a hexagonal array. Primary coolant salt is ZrF_4-KF stabilized with ZrF_2 at maximum temperature of 590°C. Secondary coolant is nitrate salt buffer. Burn-up is 120-200 GWd/t. A 750 MWt/300 MWe demonstration plant was envisaged, the SSR-W300. However, a limited market for this version is anticipated. An agreement has been signed with New Brunswick Power for initial deployment at Point Lepreau in Canada, and in March 2021 the Canadian government announced a C$50.5 million investment towards this. In April 2021 plans were confirmed for this plus a plant for recycling used Canadian nuclear fuel for it. In November 2020 the two companies were joined by ARC Canada in setting up an SMR vendor cluster there. A Canadian Nuclear Safety Commission pre-licensing vendor design review has commenced.

The company then announced the physically larger and more expensive SSR-U 'global workhorse version' of its design, with a thermal neutron spectrum running on LEU fluorides (up to 7% enriched) with graphite built into the fuel assemblies, which increases the size of the core. It runs at a higher temperature than the fast version – minimum 600°C – with ZrF_4-NaF coolant salt stabilized with ZrF_2. As well as electricity, hydrogen production is its

purpose. It is designed to be compatible with thorium breeding to U-233. It is seen as having a much larger potential market, and initial deployment in the UK in the 2030s is anticipated, with potential for replacing CCGT and coal plants.

In the SSR-Th, thorium would be in the coolant salt and the U-233 produced is progressively dissolved in bismuth at the bottom of the salt pool. This contains U-238 to denature it. Once the desired level of U-233 is achieved, the bismuth with uranium would be taken out batch-wise, and the mixed-isotope uranium is chlorinated to become fuel. If the fuel is used in a fast reactor, plutonium and actinides can be added.

Moltex has also put forward its GridReserve molten salt heat storage concept to enable the SSR-W reactor to supplement intermittent renewables, and so that the 300 MWe reactor can peak at up to 900 MWe from the heat bank of secondary coolant salt.

Mark 1 Pebble Bed FHR

This was a pre-conceptual US design completed in 2014 to evaluate the potential benefits of FHR technology. A consortium including University of California, Berkeley, Oak Ridge National Laboratory and Westinghouse designed it as a 236 MWt/100 MWe pebble-bed FHR, with annular core, operating at 700°C. It is designed for modular construction, and from 100 MWe base-load it is able to deliver 240 MWe with gas co-firing for peak loads. Fuel pebbles are 30 mm diameter, much less than gas-cooled HTRs. The project looked at how FHRs might be coupled to a Brayton combined-cycle turbine to generate power, design of a passive decay heat removal system, and the annular pebble bed core. The PB-FHR has negative void reactivity and passive decay heat removal.

Other liquid-fuel types: two-fluid breeders

Liquid fluoride thorium reactor

The liquid fluoride thorium reactor (LFTR) is a heterogeneous MSR design which breeds its U-233 fuel from a fertile blanket of lithium-beryllium fluoride (FLiBe) salts with thorium fluoride. The thorium-232 captures neutrons from the reactor core to become

protactinium-233, which decays (27-day half-life) to U-233. It may be possible to separate Pa-233 on-line and let it decay to U-233. Otherwise, newly-formed U-233 forms soluble uranium tetrafluoride (UF_4), which is converted to gaseous uranium hexafluoride (UF_6) by bubbling fluorine gas through the salt (which does not chemically affect the less-reactive thorium tetrafluoride). The volatile uranium hexafluoride is captured, reduced back to soluble UF_4 by hydrogen gas, and finally is added to the FLiBe core to serve as fissile fuel. A complication is that traces of U-232 are formed, reporting with the U-233, and having highly gamma-active decay progeny.

LFTRs can rapidly change their power output, and hence be used for load-following. Because they are expected to be inexpensive to build and operate, 100 MWe LFTRs could be used as peak and back-up reserve power units.

Flibe LFTR

Flibe Energy in the USA is studying a 40 MW two-fluid graphite-moderated thermal reactor concept based on the 1960s-'70s US molten-salt reactor programme. It uses lithium fluoride/beryllium fluoride (FLiBe) salt as its primary coolant in both circuits. Fuel is uranium-233 bred from thorium in FLiBe blanket salt. Fuel salt circulates through graphite logs. Secondary loop coolant salt is sodium-beryllium fluoride (BeF_2-NaF). A 2 MWt pilot plant is envisaged, and eventually 2225 MWt commercial plants.

Other liquid fuel types: single-fluid, thermal spectrum

Integral MSR

Canada-based Terrestrial Energy has designed the Integral MSR. This simplified MSR, based closely on the Oak Ridge MSRE, integrates the primary reactor components, including primary heat exchangers to secondary clean salt circuit, in a sealed and replaceable core vessel that has a projected life of seven years. The IMSR will operate at 600-700°C, which can support many industrial process heat applications. It operates in the thermal neutron spectrum with a hexagonal arrangement of graphite elements forming the moderator. The fuel-salt is a eutectic of low-enriched

(2-4%) uranium-235 fuel (as UF_4) and a fluoride carrier salt – likely sodium rubidium fluoride. Initially no Li or Be will be used but there is potential to change to FLiBe – at atmospheric pressure. Secondary loop coolant salt is ZrF_4-KF. Multiple pumps and six heat exchangers allow for redundancy. Emergency cooling and residual heat removal are passive. When the sealed core is replaced after seven years, it is then left for fission products to decay. Each plant would have space for two reactors, allowing seven-year changeover, with the used unit removed for off-site reprocessing when it has cooled.

The IMSR was designed in three sizes: 80 MWt (32.5 MWe), 300 MWt, and 600 MWt but the current concept is 440 MWt/195 MWe. The total levelized cost of electricity from the largest is projected to be competitive with natural gas. The smallest is designed for off-grid, remote power applications, and as a prototype. The company completed the first of three phases in the Canadian Nuclear Safety Commission's pre-licensing vendor design review in October 2018. It aims to have the first IMSRs in operation before 2030.

Transatomic TAP

Transatomic Power is a new US company partly funded by Founders Fund and aiming to develop a single-fluid MSR using very low-enriched uranium fuel (1.8%) or the entire actinide component of used LWR fuel. The TAP reactor has an efficient zirconium hydride* moderator and a LiF-based fuel salt bearing the UF_4 and actinides, hence a very compact core. The secondary coolant is FLiNaK salt (LiF-KF-NaF) to a steam generator.

* as used in TRIGA research reactors and TOPAZ and SNAP reactors for space programme.

Owing to the ZrH moderator, there are significantly more neutrons in the thermal region (less than 1 eV) compared with a graphite moderator, thereby enabling the reactor to generate power from very low-enriched uranium or used LWR fuel. The epithermal (1 eV - 1 MeV) spectrum is lower than that with graphite, but in the fast spectrum (over 1 MeV) the neutron flux is greater than with graphite moderator, and therefore contributes strongly to actinide burning. It would give up to 96% actinide burn-up. Fission products are mostly removed batch-wise and fresh fuel added. In addition to negative void and thermal coefficients, the moderator starts to fail at higher temperatures due to hydrogen loss. Decay heat removal can be by convection.

After a 20 MWt demonstration reactor, the envisaged first commercial plant will be 1250 MWt/550 MWe running at 44% thermal efficiency with 650°C in the primary loop, using a steam cycle. The overnight cost for an nth-of-a-kind 550 MWe plant, including lithium-7 inventory and online fission product removal and storage, is estimated at $2 billion with a three-year construction schedule. A version of the reactor may utilize thorium fuel.

The company had to withdraw some exaggerated claims concerning actinide burn-up made in *MIT Technology Review* in 2016. In September 2018 the company announced that it would cease operations and make its intellectual property freely available online.

Fuji MSR

The Fuji MSR is a 100-200 MWe graphite-moderated design to operate as a near-breeder with ThF_4-UF_4 fuel salt and FLiBe coolant at 700°C. It can consume plutonium and actinides, and be from 100 to 1000 MWe. Batch reprocessing. It is being developed internationally by a Japanese, Russian and US consortium: the International Thorium Molten Salt Forum (ITMSF), based in Japan. It is derived from the Oak Ridge MSBR, and several variants have been designed, including a 10 MWe mini Fuji. Thorium Tech Solutions Inc (TTS) plan to commercialise the Fuji concept, and is working on it with the Halden test reactor in Norway.

Thorcon

Martingale in the USA is designing the ThorCon MSR, which is a 250 MWe scaled-up Oak Ridge MSRE. It is a single-fluid thorium converter reactor in the thermal spectrum, graphite moderated. It uses a combination of U-233 from thorium and low-enriched U-235 from mined uranium. Fuel salt is sodium-beryllium fluoride (BeF_2-NaF) with dissolved uranium and thorium tetrafluorides (Li-7 fluoride is avoided for cost reasons). Secondary loop coolant salt is also sodium-beryllium fluoride. It operates at 700°C. There is no on-line processing – this takes place in a centralized plant at the end of the core life, with off-gassing of some fission products meanwhile. A pilot plant would be similar to the mini Fuji.

Several 550 MWt units would comprise a power station, and a 1000 MWe Thorcon plant would comprise about 200 factory- or shipyard-build modules installed below grade (30 m down). All

components are deigned to be easily and frequently replaced. For instance, every four years the entire primary loop would be changed out, returned to a centralized recycling facility, decontaminated, disassembled, inspected, and refurbished. Incipient problems would be rectified and major upgrades could be introduced without significantly disrupting power generation. The company claims generation costs of 3 to 5 c/kWh depending on scale, and is "targeting its first installations in forward-looking countries that support technology-neutral nuclear regulations and see the benefits of the license-by-test process."

Seaborg Compact Molten Salt Reactor

Seaborg Technologies in Denmark (founded 2015) has a thermal-epithermal single fluid reactor design for a 50 MWt pilot unit Compact Molten Salt Reactor (CMSR) with a view to 250 MWt commercial modular units fuelled by spent LWR fuel and thorium. Fuel salt is Li-7 fluoride initially with uranium as fluoride. Later, thorium, plutonium and minor actinides as fluorides are envisaged as fuel, hence the reactor being called a waste burner. This is pumped through the graphite column core and heat exchanger. Fission products are extracted online. Secondary coolant salt is FLiNaK, at 700°C. Spent LWR fuel would have the uranium extracted for recycle, leaving plutonium and minor actinides to become part of the MSR fuel, with thorium. The company claims very fast power ramp time. High temperature output will allow application to hydrogen production, synthentic fuels, etc.

In March 2017 the public funding agency Innovation Fund Denmark made a grant to Seaborg to "build up central elements in its long-term strategy and position itself for additional investments required to progress towards commercial maturity." This is the first Danish investment into nuclear fission research since the country introduced a ban on nuclear power in 1985. In December 2020 the American Bureau of Shipping (ABS) issued a feasibility statement regarding the reactor’s use on barges. This is the first stage in the ABS's five-phase New Technology Qualification process. Seaborg aims to deploy the first full-scale prototype power barge by 2025.

Other liquid fuel types: single-fluid, fast spectrum

Southern Co, MCFR

Southern Company Services in the USA is developing a molten chloride fast reactor (MCFR) with TerraPower, Oak Ridge National Laboratory (ORNL) – which hosts the work, the Electric Power Research Institute (EPRI) and Vanderbilt University. No details are available, and it is not certain that it is a single-fluid type. However, fuel is in the chloride salt (see section above) and as a fast reactor it can burn U-238, actinides and thorium as well as used light water reactor fuel, requiring no enrichment apart from the initial fuel load (these details from TerraPower, not Southern). It is reported to be large.

In January 2016 the US DOE awarded a Gateway for Accelerated Innovation in Nuclear (GAIN) grant to the project, worth up to $40 million. In August 2016 Southern Nuclear Operating Company signed an agreement to work with X-energy to collaborate on development and commercialization of their respective small reactor designs. With TerraPower and ORNL, X-energy is designing the Xe-100 pebble-bed HTR of 48 MWe.

Elysium MCSFR

Elysium Industries in the USA and Canada have the Molten Chloride Salt Fast Breeder Reactor (MCSFR) design with fuel in the chloride salt. It operates below grade at near atmospheric pressure. Primary fuel salt and secondary salt convey heat to steam generators at 650°C. It is designed to load-follow. A range of sizes from 125 to 3000 MWt (50 MWe to 1200 MWe) are under consideration. Used fuel from light water reactors or depleted uranium with some plutonium can fuel it though in 2020 fuel was shown as $PuCl_3$ with fission products, or 15% HALEU. Selected fission products are removed online. Passive safety includes a freeze plug. It has negative temperature and void coefficients.

Notes & references

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